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A report on "Entity Relationship (ER) Diagram"

Submitted to

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I am profoundly grateful for the contributions of all those who have supported me in this endeavor, and I sincerely hope that this ER diagram will contribute to the development of a more efficient and user-friendly voting online site.

ABSTRACT

The Entity-Relationship (ER) diagram serves as a fundamental tool in database design, providing a visual representation of the logical structure and interrelationships among various entities within a system. This report presents a comprehensive exploration of ER diagrams, detailing their significance, components, and practical applications.

Beginning with an introduction to ER diagrams, the report elucidates their role as a conceptual blueprint for designing databases, outlining the fundamental components and their relationships within the database structure. It highlights the importance of ER diagrams in providing a clear and visual representation of data relationships, aiding in understanding, documentation, and maintenance of the database structure.

A detailed examination of the symbols used in ER diagrams is provided, offering insights into the representation of entities, attributes, and relationships. Through illustrative examples and figures, the report elucidates the visual depiction of strong and weak entities, key attributes, composite attributes, multivalued attributes, and derived attributes, facilitating a deeper understanding of ER diagram components.

Moreover, the report delves into the various types of relationships depicted in ER diagrams, including one-to-one, one-to-many, many-to-one, and many-to-many relationships. Each type is elucidated with clear examples, elucidating the manner in which entities interact and establish associations within a database system.

Furthermore, the report outlines the step-by-step process for creating an ER diagram, emphasizing the importance of proper entity identification, relationship determination, attribute attachment, and removal of redundant elements. Additionally, it underscores the significance of color coding to enhance the visualization and comprehension of database structures.

In conclusion, the report presents an ER diagram of an online voting site as a practical application, showcasing the implementation of the concepts elucidated throughout the document. Through this exploration, readers gain a comprehensive understanding of ER diagrams and their pivotal role in database design and management.

1. INTRODUCTION TO ENTITY RELATIONSHIP (ER)-DIAGRAM

An Entity-Relationship (ER) diagram is a visual representation used in database design to depict the logical structure and relationships among various entities within a system. It's a powerful tool for modeling the essential components of a database and how they relate to one another.

An Entity-Relationship (ER) model serves as a conceptual representation or blueprint for designing a database. It outlines the fundamental components and their relationships within the database structure.

1.1. DEFINITION

An Entity-Relationship (ER) diagram visually represents a data model, showcasing the entities, their attributes, and the relationships between entities within a system or database. Creating an ER diagram is a structured approach to database design, necessitating a thorough analysis of all data requirements prior to implementation.

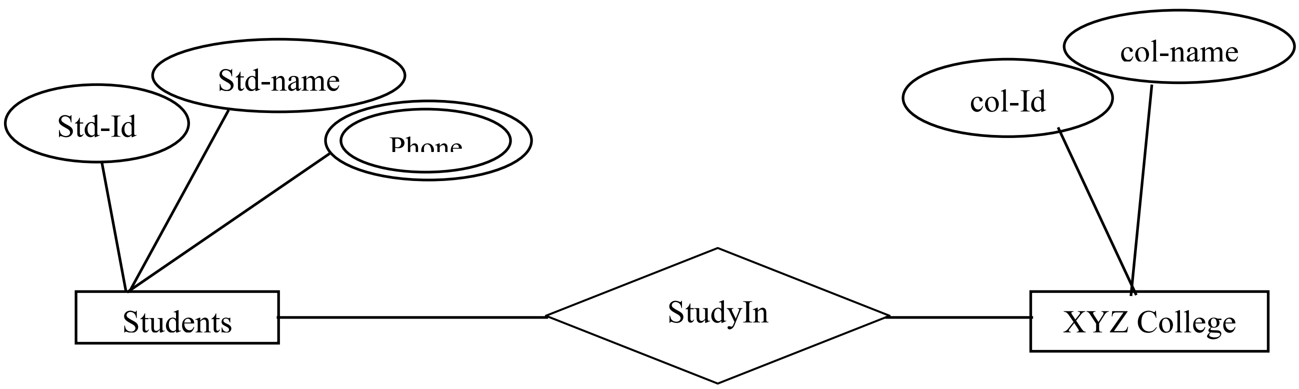

Fig 1.1: ER Diagram Introduction

1.2. SIGNIFICANCE

- ER diagrams provide a clear and visual representation of how different pieces of data relate to each other in a database.
- ER diagrams depict how different entities are linked, making it easy to understand how data is related and used.
- ER diagrams act as documentation of the database structure, aiding future maintenance and updates.
- ER diagrams help ensure that the database design meets the requirements of the system it supports.

1.3. SYMBOLS USED IN ER DIAGRAM

ER COMPONENT	DESCRIPTION	SYMBOLS
Strong Entity	Rectangle box	
Weak Entity	Double Rectangle	
Relationship	Rhombus	
Weak entity	Double lined rhombus	
Attributes	Ellipse	
Key Attribute	Underline the attribute name inside Ellipse	Key_Attribute
Derived Attribute	Dotted Ellipse	
Multivalued Attribute	Double ellipse	

Fig 1.3.: Symbol used in ER-Diagram

2. COMPONENT OF ER DIAGRAM

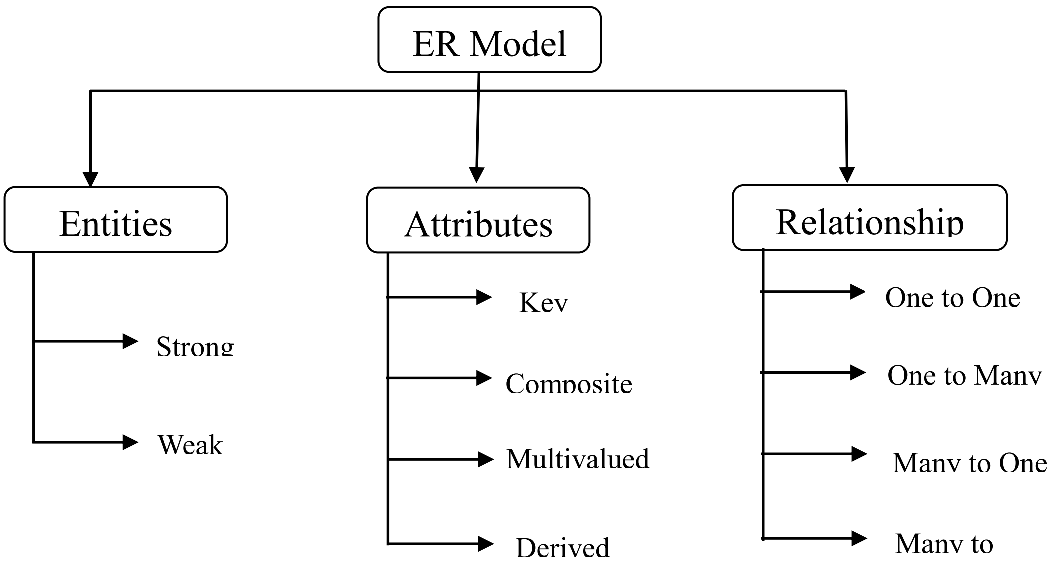

Fig 2: Component of ER Diagram

2.1. ENTITY

An entity is a fundamental component in an ER diagram, representing a distinct object, concept, or thing in the real world that is relevant to the system being modeled. It can have attributes (properties or characteristics) that describe it. Entities is represented as a rectangle with rounded corner.

2.1.1. Strong Entity:

A strong entity is an entity set that possesses a key attribute or a combination of attributes that uniquely identifies each instance within the set. It corresponds to a table in a database that contains a primary key column. Strong entities are depicted by a single rectangular shape in an Entity-Relationship (ER) diagram.

When representing the association between two strong entities, a single diamond symbol is utilized. This diamond symbol connects the two rectangular shapes representing the strong entities through lines, effectively illustrating the relationship that exists between them.

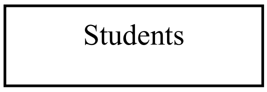

Fig 2.1.1: Symbol of Strong Entity

2.1.2. Weak Entity:

A weak entity set is an entity set that does not possess a primary key attribute or a combination of attributes that can uniquely identify its individual entities. Unlike strong entity sets, which have a self-contained primary key, weak entity sets rely on the primary key of a related strong entity set for unique identification. However, weak entity sets do contain a partial key, known as a discriminator, which can identify groups of entities within the set but cannot distinguish between individual entities.

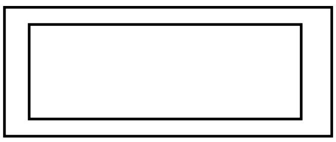

Fig 2.1.2: Symbol of weak Entity

2.2. ATTRIBUTES

An attributes describe the property of an entity. It is represented as Oval in ER Diagram. Types are described below: -

2.2.1. Key Attributes

A key attribute possesses the ability to uniquely identify each individual entity within an entity set. In an Entity-Relationship (ER) diagram, the visual representation of a key attribute is achieved by underlining the text corresponding to that specific attribute. For instance, “Employee ID” could be a key attribute for the “Employee” entity.

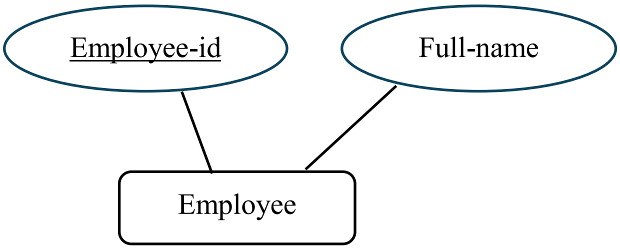

Fig 2.2.1: Key Attributes

2.2.2. Composite Attributes

These attributes can be divided into smaller sub-parts, representing more basic attributes with independent meanings. For instance, a “Date” attribute can be divided into “year” and “month”.

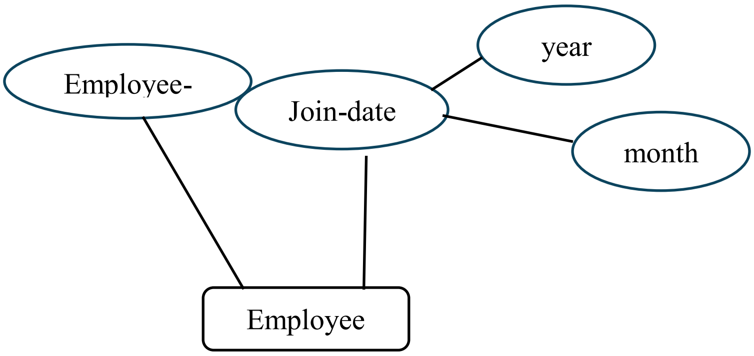

Fig 2.2.2: Composite Attributes

2.2.3. Multivalued Attributes

Some attributes can possess over one value, those attributes are called multivalued attributes. The double oval shape is used to represent a multivalued attribute. For instance, a “language spoken” attribute might be multi-valued if employees can have more than one phone number.

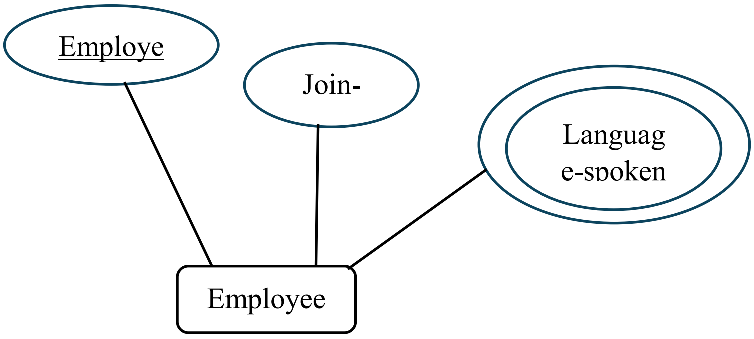

Fig 2.2.3: Multivalued Attributes

2.2.4. Derived Attributes:

Derived Attributes: A derived attribute is one whose value is dynamic and derived from another attribute. It is represented by dashed oval in an ER Diagram. For example, an employee’s years can be derived from their “Join Date”.

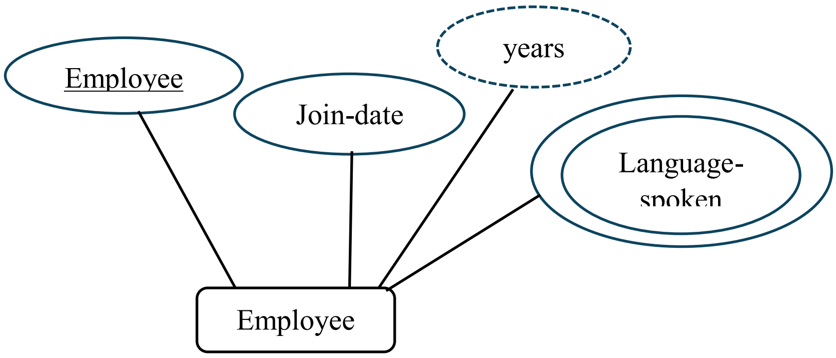

Fig 2.2.4: Derived Attributes

3. Relationship

A Relationship is a logical link between two or more entities, depicting how the entities interact with each other. Relationships help to define the associations and dependencies between entities in a database.

There are four types of relationship: -

3.2. One to One Relationship: When a single instance of an entity is associated with a single instance of another entity then it is called one to one relationship.

For examples, a citizen has only one citizenship card and an identification card is given to one person.

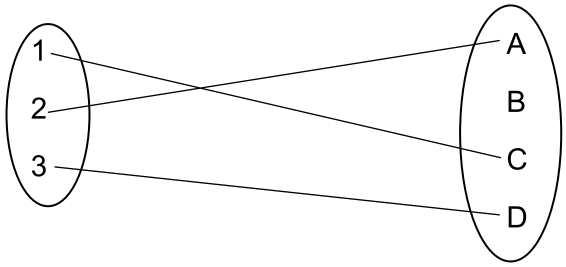

Fig 3.1.1. : One to One Relationship (Mapping)

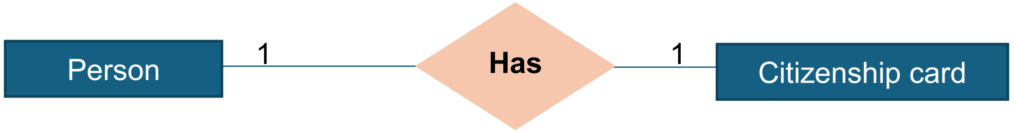

Fig 3.1.2.: Notation used in ER Diagram for One to One Relationship

3.3. One to Many Relationship

This is a relationship where a single instance of an entity A can be associated with multiple instances of entity B, but each instance of entity B can be associated with only one instance of entity A. For example, a 'Department' entity can have a one-to-many relationship with an 'Employee' entity, representing that one department can have many employees, but each employee belongs to one department.

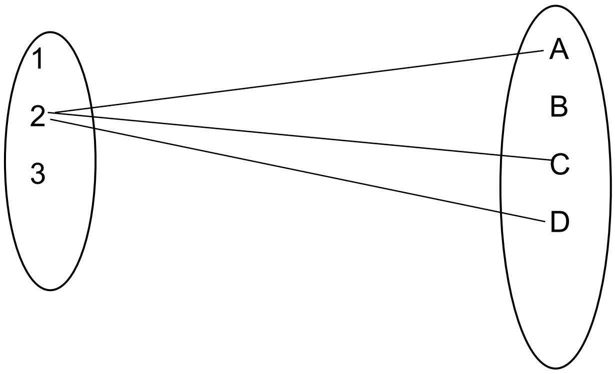

Fig 3.2.1.: One to Many Relationship (Mapping)

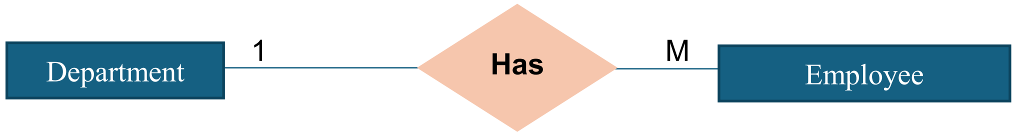

Fig 3.2.2.: Notation used in ER Diagram for One to Many Relationship

3.4. Many to One Relationship

When more than one instance of the entity on the left, and only one instance of an entity on the right associates with the relationship then it is known as many to one relationship. For Example many student can enroll one same course.

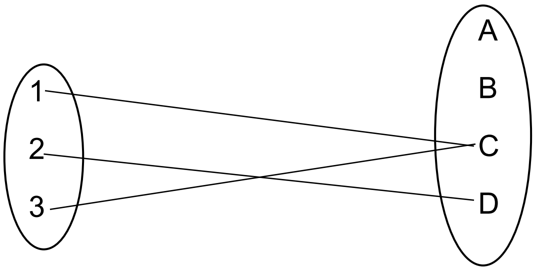

Fig 3.3.1.: Many to One Relationship (Mapping)

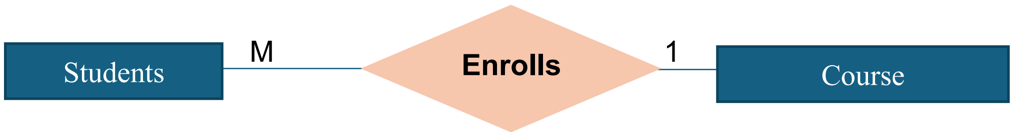

Fig 3.3.2.: Notation used in ER Diagram for Many to One Relationship

3.5. Many to Many Relationship

When more than one instances of an entity is associated with more than one instances of another entity then it is called many to many relationship. For example, an 'Author' entity can have a many-to-many relationship with a 'Book' entity, indicating that one author can write many books, and one book can have many authors.

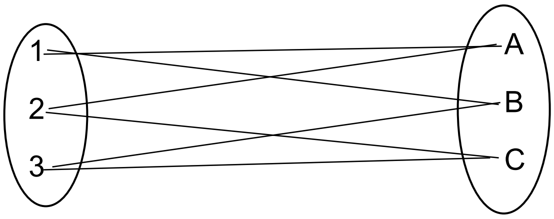

Fig 3.4.1.: Many to Many Relationship (Mapping)

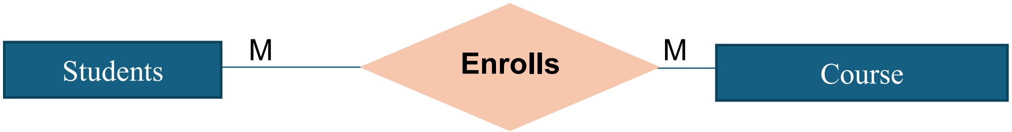

Fig 3.4.2.: Notation used in ER Diagram for Many to Many Relationship

4. How to Draw ER Diagram?

The process to draw ER Diagram is given steps followed: -.

- **Identify Entities:** Begin by identifying the main entities in your database. These are typically the nouns in your problem domain.
- **Identify Attributes:** For each entity, identify the attributes or properties that describe it. These are typically the adjectives or characteristics of the entities.
- **Determine Relationships:** Identify the relationships between the entities. Determine the cardinality (e.g., one-to-one, one-to-many, many-to-many) and participation constraints (e.g., mandatory or optional) for each relationship.
- **Draw Entities and Attributes:** Draw rectangles to represent entities and ovals to represent attributes within those rectangles. Connect attributes to their respective entities.
- **Draw Relationships:** Draw lines between related entities to represent relationships. Use diamonds to represent relationship sets. Label the lines with appropriate cardinality ratios (e.g., 1:1, 1:N, M:N).
- **Add Primary and Foreign Keys:** Identify the primary key of each entity and underline it. For relationships, identify foreign keys and connect them to the respective entities with dashed lines.
- **Refine and Review:** Refine the diagram by reviewing it for accuracy and completeness. Ensure that all entities, attributes, and relationships are properly represented.

5. ER Diagram of Library Management System.

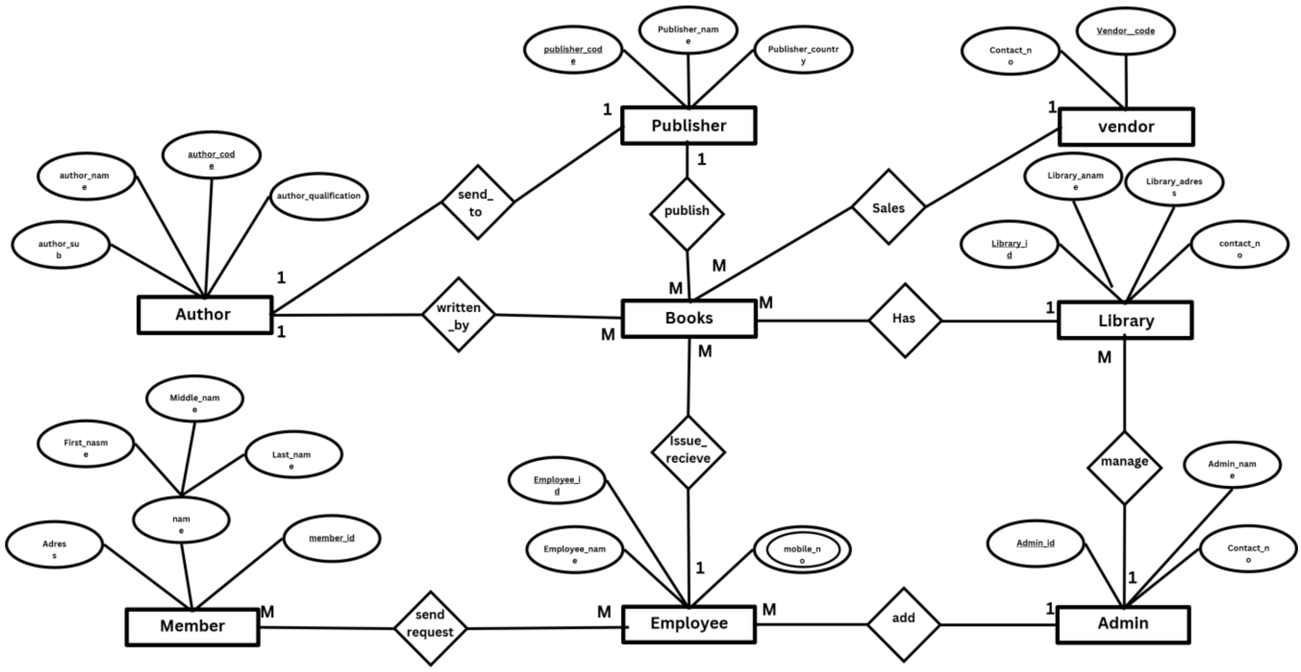

Fig 5: ER Diagram of Library Management System

The ER diagram in the image illustrates a comprehensive library management system with detailed relationships among various entities, such as Author, Publisher, Vendor, Library, Member, Books, Employee, and Admin. Each entity contains unique attributes to identify and describe its role within the system. For instance, Authors have attributes like author_code and author_name, representing their contribution to the system by writing multiple books, establishing a many-to-many relationship with the Books entity. Publishers, identified by attributes such as publisher_code and publisher_name, maintain a one-to-many relationship with books, as each publisher can release numerous books, but each book is published by a single publisher.

Vendors supply books to various Libraries, which are managed by Admins. The vendor-library relationship reflects a one-to-many connection, where each vendor is associated with multiple libraries, but each library has one vendor managing it. Admins oversee the libraries and have the capability to manage multiple libraries, forming a one-to-many relationship.

Members interact with the system by requesting books, and Employees handle these requests and manage the issuing and receiving of books, forming a many-to-many relationship between members and employees, as well as between employees and books. This setup allows for efficient handling of book circulation within the library.

Employees also manage the operational aspects of book distribution and return, reinforcing their pivotal role in ensuring smooth transactions within the system. The ER diagram effectively showcases the complex yet well-organized relationships and interactions within the library, emphasizing the connectivity and interdependence of various entities in maintaining an effective library system.The Result entity, derived from Votes, represents the conclusion of the voting process. It establishes a many-to-one relationship with Candidates, permitting multiple Candidates to access and interpret the election outcome.

In summary, the ER diagram demonstrates a well-structured voting system where Admins manage Users, who can take on roles as either Candidates or Voters. Candidates engage with Voters through dynamic interactions facilitated by the many-to-many relationship. The creation of Votes by Admins and the computation of Results ensure an efficient and transparent election process, promoting trust and credibility within the system.

6. CONCLUSION

This This report explores the importance and construction of Entity-Relationship (ER) diagrams, particularly for a library management system. ER diagrams are crucial in database design, offering a clear visual representation of the interactions and relationships between different data entities within the system.

We began by introducing the purpose and key symbols of ER diagrams, detailing components like strong and weak entities, various attributes, and the relationships between entities. A step-by-step guide was provided to create accurate ER diagrams, enhancing understanding.

A practical example was presented through an ER diagram for a library management system, illustrating the interactions between entities such as Author, Publisher, Vendor, Library, Member, Books, Employee, and Admin. This example mapped out the system's structure, ensuring a comprehensive understanding of its database.

In summary, ER diagrams are essential tools for designing efficient databases in library management. They help visualize data relationships, facilitating easier development, documentation, and maintenance of the system. This report emphasizes both the theoretical and practical significance of ER diagrams, underscoring their role in creating well-structured and effective database designs for managing libraries.